

Recall

- $\mathbb{H} = \{q = a + bi + cj + dk \mid a, b, c, d \in \mathbb{R}\}$ is a $\mathbb{R}$ -vs. with $\dim = 4$.
- It is also a ring with subring $\mathbb{R}$.
- $\mathbb{R}$ acts on $\mathbb{H}$ by multiplication $:=$ the $\mathbb{R}$ -multiplication on $\mathbb{R}$ -vs.

Moreover $\mathbb{C} \hookrightarrow \mathbb{H}$ as a subring. Injectivity is obvious. Properties of $+$, 1 , $\cdot$ are preserved.
 $\implies \mathbb{H}$ becomes a $\mathbb{C}$ -vector space.

- $\dim_{\mathbb{C}} \mathbb{H} = 2$: every $q \in \mathbb{H}$ has unique expression $q = z + wj$,
 $\because a + bi + cj + dk = (a + bi) + (c + di)j$.

$\mathbb{H}$ is not commutative by definition, its center $Z(\mathbb{H})$ equals $\mathbb{R}$.

$\because$ Multiplication $\mathbb{H} \times \mathbb{H} \rightarrow \mathbb{H}$ is $\mathbb{R}$ -bilinear, hence $z \in Z(\mathbb{H}) \iff \forall x \in \{1, i, j, k\}, zx = xz$.

Next: show $\mathbb{H}$ is a division ring, i.e. $\mathbb{H}^\times = \mathbb{H} \setminus \{0\}$.

Definition For $q = a + bi + cj + dk \in \mathbb{H}$, define its

- **conjugate** $\bar{q} := a - bi - cj - dk$,
- **trace** $\text{Tr}(q) := q + \bar{q} = 2a$,
- **norm** $N(q) := q\bar{q}$.

Properties

- $q \mapsto \bar{q}$ is $\mathbb{R}$ -linear, and $\overline{\bar{q}} = q$.
- $\overline{q_1 q_2} = \bar{q}_2 \cdot \bar{q}_1$, $\because$ both sides are $\mathbb{R}$ -linear, suffices to check $q_1, q_2 \in \{1, i, j, k\}$.
- $\text{Tr}(q)$ is $\mathbb{R}$ -linear.
- (properties of norm)
 - $N(1) = 1$,
 - $N(q) \in \mathbb{R}$, $\because \overline{q \cdot \bar{q}} = \bar{\bar{q}} \cdot \bar{q} = q\bar{q}$.
 - $N(q_1 q_2) = N(q_1)N(q_2)$, $\because q_1 q_2 \overline{q_1 q_2} = q_1 q_2 \bar{q}_2 \cdot \bar{q}_1 = N(q_2)N(q_1)$.
 - $N(a + bi + cj + dk) = a^2 + b^2 + c^2 + d^2 \geq 0$.

Theorem $\mathbb{H}^\times = \mathbb{H} \setminus \{0\}$. In fact, $q \neq 0 \implies q^{-1} = N(q)^{-1}\bar{q}$.

Proof $q \neq 0 \implies N(q) > 0$.

$$q \cdot N(q)^{-1}\bar{q} = N(q)^{-1}q\bar{q} = 1,$$
$$N(q)^{-1}\bar{q} \cdot q = 1.$$

□

Remark Frobenius proved that if D : division ring + $\mathbb{R}$ -vs. structure s.t. multiplication $D \times D \rightarrow D$ is $\mathbb{R}$ -bilinear, $\dim_{\mathbb{R}} D < \infty$, then $\exists$ isomorphism of $\mathbb{R}$ -vs. and ring $D \simeq \mathbb{R}, \mathbb{C}$ or $\mathbb{H}$.

$\exists$ embedding of rings $\Phi : \mathbb{H} \hookrightarrow M_{2 \times 2}(\mathbb{C}), z + wj \mapsto \begin{pmatrix} z & -\bar{w} \\ w & \bar{z} \end{pmatrix}, z, w \in \mathbb{C}$. Injectivity is clear,
 $\Phi(1) = 1$, $\mathbb{R}$ -linear (preserves $+$). To show $\Phi(q_1 q_2) = \Phi(q_1)\Phi(q_2)$, use $ji = -ij$ and $j^2 = -1$ to compute.

□

In this sense,

- $\text{Tr of } \mathbb{H} \longleftrightarrow \text{Tr in } M_{2 \times 2}(\mathbb{C})$,
- $\text{norm of } \mathbb{H} \longleftrightarrow \det \text{ in } M_{2 \times 2}(\mathbb{C})$.

Under this projection,

$$i \mapsto \begin{pmatrix} i & \\ & -i \end{pmatrix}, \quad j \mapsto \begin{pmatrix} & 1 \\ -1 & \end{pmatrix}, \quad k \mapsto \begin{pmatrix} & -i \\ i & \end{pmatrix}.$$

These matrices are (up to $\pm$) called Pauli matrices.

Similarly $\mathbb{C} \hookrightarrow M_{2 \times 2}(\mathbb{R}), a + bi \mapsto \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$.

§ Quaternions and rotation in $\mathbb{R}^3$

$\mathbb{R}^3 \simeq \mathbb{H}_0 := \{q \in \mathbb{H} \mid q = -\bar{q}\} = \{q \mid \text{Tr}(q) = 0\} = \{ai + bj + ck \mid a, b, c \in \mathbb{R}\}$.

If we transport the std IP on $\mathbb{R}^3$ to $\mathbb{H}_0$, then $\forall q \in \mathbb{H}_0, \|q\|^2 = N(q)$.

Lemma Let $x \in \mathbb{H}^\times$.

- $N(xqx^{-1}) = N(q), \forall q \in \mathbb{H}$,
- We have an orthogonal transformation $R_x : q \mapsto xqx^{-1}$ on $\mathbb{H}_0$,
- $\det R_x = 1$.

Proof

- $N(x)N(x^{-1}) = N(xx^{-1}) = N(1) = 1$. Hence $N(xqx^{-1}) = N(x)N(q)N(x^{-1}) = N(q)$.

Similarly $\overline{x^{-1} \cdot \bar{x}} = \overline{x \cdot x^{-1}} = 1 = \overline{x^{-1} \cdot x} = \bar{x} \cdot \overline{x^{-1}} \implies \overline{x^{-1}} = \bar{x}^{-1}$.

- Let $q \in \mathbb{H}_0$ and assume $N(x) = 1 (x^{-1} = \bar{x})$.

$\overline{xqx^{-1}} = \overline{x^{-1} \cdot \bar{q} \cdot \bar{x}} = x\bar{q} \cdot \bar{x} = -xq\bar{x} \implies xq\bar{x} \in \mathbb{H}_0$.

$\mathbb{R}$ -linear is given by definition. R_x is also isometry by (1), hence case $N(x) = 1$ is proved.

In general $x = \sqrt{N(x)} \cdot y, N(y) = 1, xqx^{-1} = yqy^{-1}$.

- Identify $\mathbb{H}$ with $\mathbb{R}^4$ and $\mathbb{H}_0$ with $\mathbb{R}^3 \implies R_x$ identified with a 3×3 orthogonal matrix, hence $\det T \in \{\pm 1\}$. All entries of the matrix are continuous functions in $x \in \mathbb{H}^\times \simeq \mathbb{R}^4 \setminus \{0\}$, i.e. $\mathbb{R}^4 \setminus \{0\} \rightarrow \{\pm 1\}, x \mapsto \det R_x$ is continuous. Note that $x = 1$ gives $\det R_1 = \det 1_{3 \times 3} = 1$. Now $\mathbb{R}^4 \setminus \{0\}$ is connected and $\{\pm 1\}$ is discrete, hence it is a constant function. $\det R_x \equiv 1$. $\square$

Observations

- $R_{xy} = R_x R_y, \because (xy)q(xy)^{-1} = x(yqy^{-1})x^{-1} = (R_x R_y)(q) \implies R_x^{-1} = R_{x^{-1}}$.
- $R_{tx} = R_x, \forall t \in \mathbb{R}^\times$.

From above we showed that R_x acts as a rotation in $\mathbb{H}_0 \simeq \mathbb{R}^3$ for all $x \in \mathbb{H}^\times$.

Question: does $\forall$ rotation arise in this way? (if so, may assume $N(x) = 1$ by $/\sqrt{N(x)}$.)

Theorem Let T : rotation in $\mathbb{H}_0$. $\exists x \in \mathbb{H}, N(x) = 1$ s.t. $T = R_x$ and x is unique up to $\pm$.

Proof Existence: Identify $\mathbb{H}_0$ with $\mathbb{R}^3$. Euler angles: $\exists \varphi, \theta, \psi \in \mathbb{R}$,

$$T = R_{e_3}(\psi)R_{e_2}(\theta)R_{e_3}(\varphi).$$

Hence reduce to the case $T = R_{e_\ell}(\psi), \ell \in \{1, 2, 3\}, \psi \in \mathbb{R}$.

For $\ell = 1$: take $x = \cos \theta + \sin \theta i \in \mathbb{H}^\times, N(x) = 1, x^{-1} = \bar{x} = \cos \theta - \sin \theta i$.

$xix^{-1} = i$ by working in $\mathbb{C}$,

$$xjx^{-1} = (\cos \theta + \sin \theta i)j(\cos \theta - \sin \theta i) = \cos 2\theta j + \sin 2\theta k,$$

$$xkx^{-1} = (\cos \theta + \sin \theta i)k(\cos \theta - \sin \theta i) = \cos 2\theta k - \sin 2\theta j,$$

In ONB (i, j, k) , R_x has matrix $\begin{pmatrix} 1 & & \\ \cos 2\theta & -\sin 2\theta & \\ \sin 2\theta & \cos 2\theta & \end{pmatrix} = R_{e_1}(2\theta)$.

Similarly,

$$x = \cos \theta + \sin \theta j \implies R_x = R_{e_2}(\pm 2\theta),$$

$$x = \cos \theta + \sin \theta k \implies R_x = R_{e_3}(\pm 2\theta).$$

Uniqueness: If $x, y \in \mathbb{H}$, $N(x) = N(y) = 1$, $R_x = R_y$, then

$$\text{id} = R_x R_y^{-1} = R_{xy^{-1}}.$$

Remains to show: $N(x) = 1, R_x = \text{id} \implies x \in \{\pm 1\}$.

$$R_x = \text{id} \implies xqx^{-1} = q, \forall q \in \mathbb{H}_0.$$

Since R_x is $\mathbb{R}$ -bilinear, we can generalize to $q \in \mathbb{H}$. Hence $xq = qx, x \in Z(\mathbb{H}) \cap \{N(x) = 1\}$. $\square$

More precisely, let $u \in \mathbb{H}_0, N(u) = 1$. $R_u(\theta)$ (definition same as the $\mathbb{R}^3$ case) equals R_x where

$$x = \cos \theta + \sin \theta \cdot u.$$